

VOLKEL AP, Netherlands

WMO: 063750

Lat: 51.6586N

Lon: 5.7067E

Elev: 72

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	22.8	13.2	10.8	22.0	17.4	13.3	24.5	28.7	48.7	25.6	48.8	6.8	70	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	86.0	68.1	81.8	66.9	78.2	65.1	70.2	81.4	68.4	78.6	66.7	75.6	7.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	97.8	74.4	64.8	92.1	72.4	63.3	87.3	70.5	34.2	81.6	32.7	78.5	31.3	75.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.1	17.6	DB	12.8	93.2	6.0	3.5	8.5	95.7	5.0	97.7	1.7	99.7	-2.7	102.2	(4)
(5)			WB	12.2	74.1	5.9	1.8	8.0	75.4	4.6	76.4	1.3	77.4	-2.9	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	37.3	38.6	43.2	49.5	56.1	61.3	65.2	64.2	58.8	51.8	44.1	38.6		
	DBStd	11.79	7.73	7.28	6.31	6.65	6.38	5.80	5.85	5.20	5.04	6.17	6.43	7.75		
	HDD50	1649	394	321	222	90	16	0	0	0	1	53	195	357		
	HDD65	5365	858	739	674	466	285	141	70	79	197	411	627	818		
	CDD50	1937	1	2	13	74	204	340	471	439	264	107	18	4		
	CDD65	180	0	0	0	1	8	31	76	53	10	1	0	0		
	CDH74	1863	0	0	0	28	122	342	761	511	94	5	0	0		
	CDH80	538	0	0	0	2	18	88	253	161	16	0	0	0		
	WSAvg	8.0	9.7	9.3	9.0	7.7	7.5	6.9	6.8	6.5	6.8	7.9	8.2	9.5		
	PrecAvg	31.5	2.7	2.6	2.0	1.7	2.4	2.7	3.1	3.1	2.5	2.6	2.9	3.1		
(15) Precipitation	PrecMax	40.3	5.2	5.4	4.5	3.2	4.5	7.0	5.8	6.5	6.4	7.4	4.9	6.1		
	PrecMin	23.9	0.2	0.6	0.2	0.0	0.7	0.6	1.0	0.8	0.3	0.9	0.6	1.0		
	PrecStd	4.0	1.3	1.3	1.1	0.8	1.1	1.4	1.4	1.7	1.6	1.2	1.1	1.3		
	PrecStd	4.0	1.3	1.3	1.1	0.8	1.1	1.4	1.4	1.7	1.6	1.2	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	58.1	66.3	78.1	82.9	88.2	93.0	90.4	82.2	73.0	62.2	56.6		
		MCWB	51.9	49.9	54.7	58.7	65.6	69.3	69.4	69.1	67.2	63.6	57.5	53.3		
	2%	DB	52.5	53.6	60.9	71.5	77.4	81.9	86.4	84.9	76.3	67.2	58.2	54.3		
		MCWB	50.0	49.1	51.8	57.1	61.8	66.3	68.6	68.0	65.1	60.7	55.0	51.9		
	5%	DB	50.1	51.3	56.5	66.4	73.6	77.7	81.9	79.9	71.9	63.6	55.8	51.9		
		MCWB	47.7	47.9	50.0	54.9	60.2	64.8	67.4	66.7	63.1	58.9	53.1	49.6		
	10%	DB	47.7	49.0	53.4	61.8	69.3	73.5	77.8	75.3	68.1	61.0	53.4	49.4		
		MCWB	45.6	46.1	48.5	53.0	58.7	62.6	65.6	64.3	60.9	57.3	51.0	47.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.0	55.8	61.4	67.6	71.7	73.2	72.2	68.8	64.1	57.8	54.4		
		MCDB	54.4	55.4	62.9	73.5	78.9	83.1	85.7	85.2	78.2	70.6	60.3	55.7		
	2%	WB	50.3	50.0	53.1	58.3	64.5	68.4	70.6	69.7	66.2	61.5	55.6	52.2		
		MCDB	52.2	52.6	58.5	68.4	73.9	78.1	81.9	80.6	74.0	65.8	57.9	54.0		
	5%	WB	47.9	48.3	51.1	56.0	61.6	66.1	68.7	67.7	64.1	59.6	53.5	50.0		
		MCDB	49.9	50.8	55.4	64.6	69.8	74.8	79.3	77.3	70.1	63.0	55.5	51.7		
	10%	WB	45.8	46.3	49.1	53.9	59.5	63.9	66.7	65.8	62.2	57.8	51.2	47.6		
		MCDB	47.5	48.9	52.9	60.7	67.5	71.7	75.9	73.6	66.8	60.7	53.0	49.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	11.3	14.6	18.3	18.6	18.6	18.4	18.6	17.0	13.8	10.5	8.9		
		MCDBR	10.2	13.3	21.0	26.4	25.8	25.7	26.1	26.3	22.0	16.8	11.6	9.3		
		MCWBR	9.0	9.7	13.3	14.0	12.6	12.3	11.1	11.7	11.9	11.0	9.2	8.3		
	5% WB	MCDBR	10.0	11.4	17.7	22.6	21.8	21.8	22.6	23.1	19.1	15.5	11.0	9.1		
		MCWBR	9.1	9.1	11.8	12.8	11.9	11.6	10.8	11.7	11.2	10.9	9.3	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.335	0.358	0.390	0.422	0.428	0.425	0.435	0.425	0.398	0.371	0.347	0.330		
	taud		2.347	2.316	2.233	2.179	2.207	2.232	2.212	2.260	2.324	2.379	2.356	2.323		
	Ebn at Noon		210	238	252	258	262	264	259	254	247	231	202	189		
	Edh at Noon		20	27	36	42	43	42	42	38	32	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		233	448	818	1316	1588	1710	1612	1370	1004	587	272	177		
	RadStd		29	86	114	140	129	135	191	120	96	49	28	22		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page